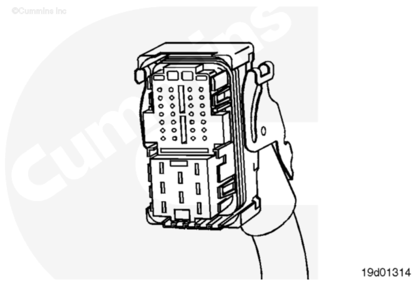
Trainings

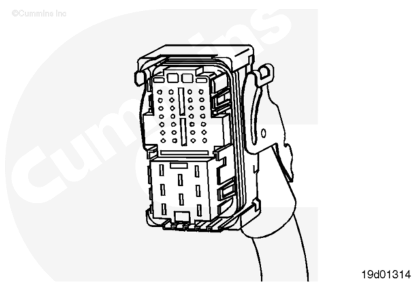
Pin Replacement

The connector is used to attach the appropriate harness to the ECM.



Remove the connector shell by slightly bending the connector shell (black) away from the two tangs that hold the shell to the ECM connector (red).

Before pins can be removed, they **must** be unlocked. Slide the purple tabs on the edges of the connector sideways at the same time. When unlocked, the purple tab will align with a slot, making the entire length of the purple tab visible.

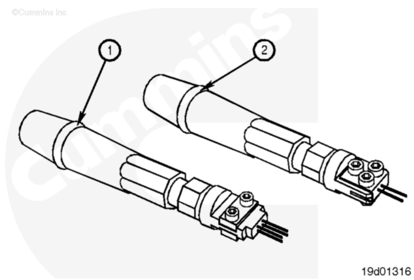


Use Bosch® extraction tool (2), Part Number 3164093, (small terminals), or use Bosch® extraction tool (1), Part Number 3164091 (large terminals), over the wire to remove a pin from the connector.

Replace one contact wire at a time. If more than one wire needs replaced, attach a lettered tag to each wire removed.

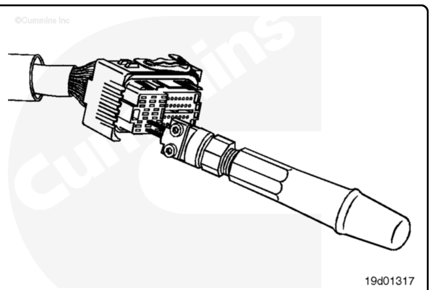
Refer to the wiring diagram in Section E for pin locations.

Refer to the appropriate wiring repair kit in the service tools table in the front of Section 19 for the correct repair wire.

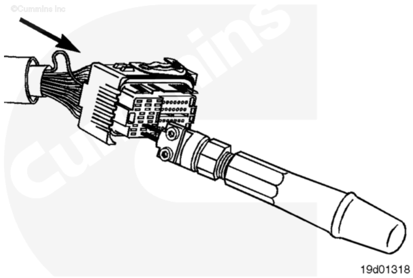


Insert the pin extractor tool into the unlocking holes in the connector.

Do **not** push the tool all the way into the connector.

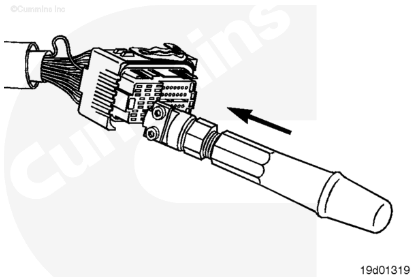


Push the corresponding wire toward the pin extraction tool.



19d01318

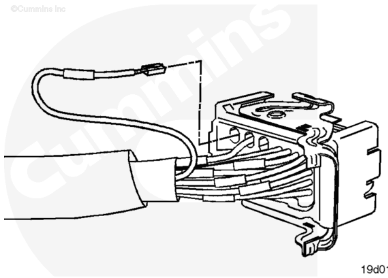
Press the pin extraction tool all the way into the connector.



19d01319

⚠ CAUTION ⚠

If the wire is difficult to remove, do not pull hard on the wire; otherwise, the locking tang of the wire terminal will stick or the terminal will pull off the wire and remain in the connector.



19d01320

Carefully pull the wire out of the connector. If it is difficult to remove, repeat the entire procedure.

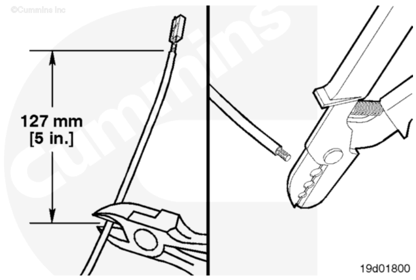
Note : The repair wire is 127 mm [5 in] long.

Use wire cutters to cut 127 mm [5 in] of the wire and pin.

Use wire crimping tool, Part Number 3822930, to remove 6 mm [¼ in] of insulation from the wire.

Refer to the appropriate wiring repair kit in the service tools table in the front of Section 19 for the correct repair wire.

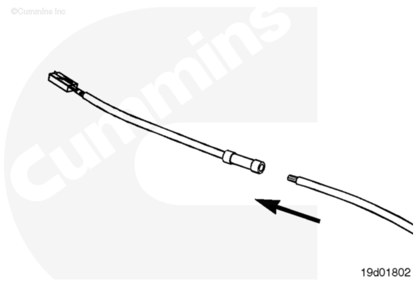
Before installing the new repair wire, perform a test fit to make sure the wire is the correct size.



19d01800

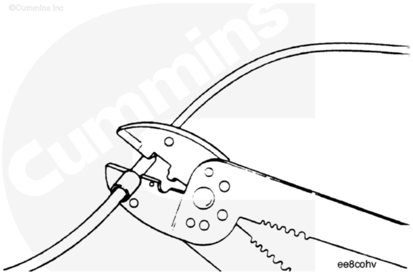
Install the repair wire on the bare wire.

Make sure the bare wire extends into the splice connector.



19d01802

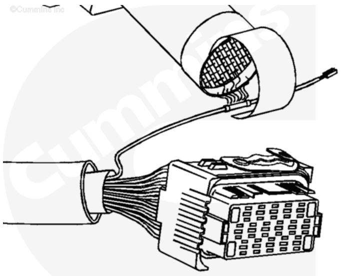
Use wiring crimping tool, Part Number 3822930, to crimp the repair wire onto the bare wire.



ee8cohv

Use heat gun, Part Number 3822860, to heat the shrink tubing around the wire.

The tubing will shrink and make the connection waterproof.



19d01803

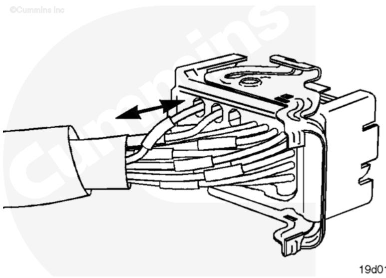
The wire terminal has locating pins that **only** allow it to be installed in a certain orientation.

Insert the wire from the backside of the connector.

Push the wire into the connector.

Pull the wire gently to make sure it is locked into the connector.

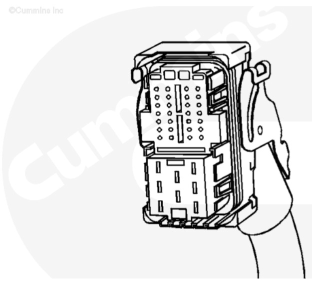
Note : If the wire's locking tang has not latched, then remove the wire, pry the tang away from the terminal, and repeat this step.



19d01321

Connector Replacement

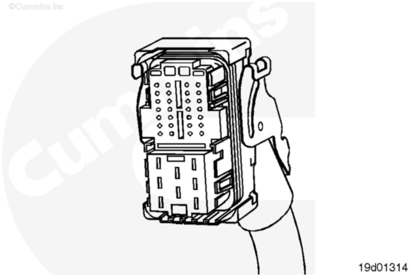
This connector is used to attach the engine harness to the ECM.



19d01314

Remove the connector shell by slightly bending the connector shell (black) away from the two tangs that hold the shell to the ECM connector (red).

Before pins can be removed, they **must** be unlocked. Slide the purple tabs on the edges of the connector sideways at the same time. When unlocked, the purple tab will align with a slot, making the entire length of the purple tab visible.

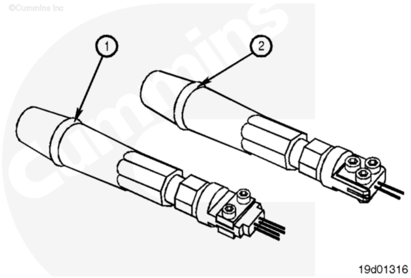


Use Bosch® extraction tool (2), Part Number 3164093, (small terminals), or use Bosch® extraction tool (1), Part Number 3164091 (large terminals), over the wire to remove a pin from the connector.

Remove one contact wire at a time. If more than one wire needs replaced, attach a lettered tag to each wire removed.

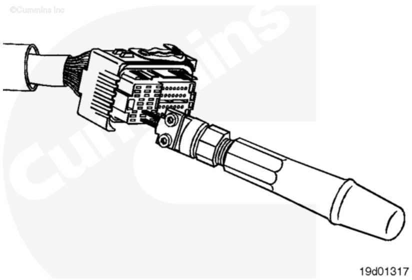
Refer to the wiring diagram in Section E for pin locations.

Refer to the appropriate wiring repair kit in the service tools table in the front of Section 19 for the correct repair wire.

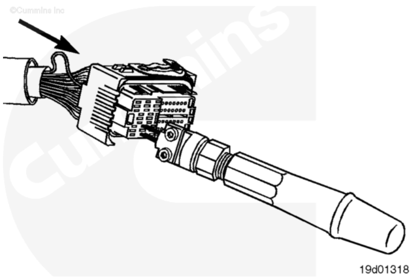


Insert the pin extraction tool into the unlocking holes in the connector.

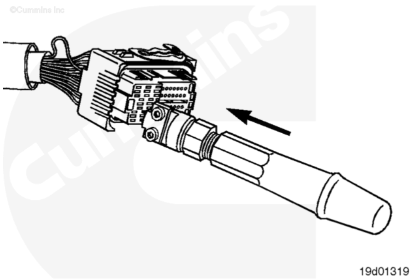
Do **not** push the tool all the way into the connector.



Push the corresponding wire toward the pin extraction tool.

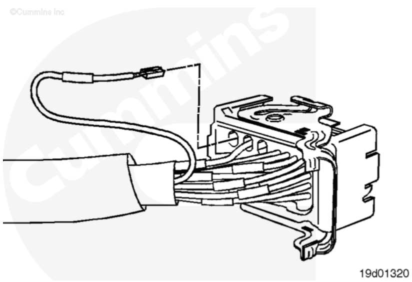


Press the pin extraction tool all the way into the connector.



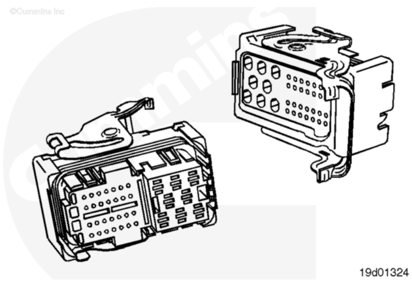
⚠ CAUTION ⚠

If the wire is difficult to remove, do not pull hard on the wire; otherwise, the locking tang of the wire terminal will stick or the terminal will pull off the wire and remain in the connector.



Carefully pull the wire out of the connector and record the hole from which the pin is removed. If it is difficult to remove, repeat the entire procedure.

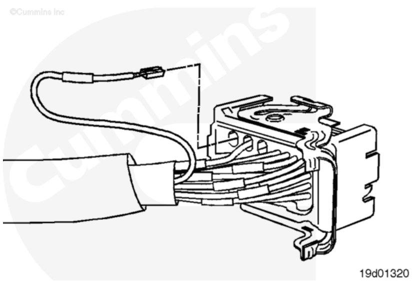
Before installing the new connector, perform a test fit to make sure the connector is keyed correctly.



Refer to the appropriate wiring repair kit in the service tools table in the front of Section 19 for the correct repair connector.

Refer to the wiring diagram in Section E for pin locations.

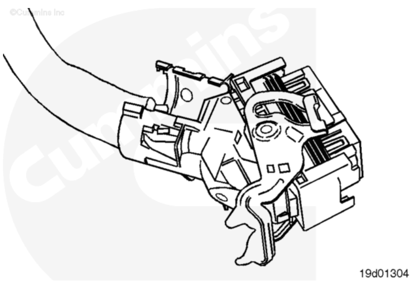
Insert the pins into the correct hole of the replacement connector.



Each pin **must** click into place and hold the wires in the connector.

Pull each wire gently to make sure it is seated in the connector.

Replace the connector shell by inserting the hinge of the connector shell (black) into the hinge of the connector (red).



Close the connector shell onto the connector and wiring harness by pressing it on the tang of the connector until a click is heard.